David Mossakowski

Engineering Manager | Product-Minded Rapid Domain Expert
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Professional Profile

Hands-on Engineering Manager with 20+ years of experience delivering high-performance systems across US Capital Markets and Retail R&D. I am an innovative problem-solver who excels at proposing non-standard solutions and maximizing the latent potential of existing infrastructure—often finding ways to advance and repurpose available technology to its full capacity. I learn quickly, thrive on complex challenges, and have a proven track record of deep-diving into diverse domains—from high-frequency trading algorithms and niche kdb+ dataseries platforms to modern AI-driven SDKs and observability systems—to deliver robust, enterprise-grade results.

Core Competencies

- Leadership: Geographically Distributed Teams, Agile/Scrum, Technical Product Ownership.
- AI & Data: RAG Architecture, Vector DBs (FAISS/Oracle), LLM Benchmarking, Prompt Engineering.
- Architecture: Distributed Systems, Microservices, API Design (REST/FIX/FpML), SDK Development.
- Specializations: Observability (OTel), Fintech/High-Frequency Trading, AI Agents, Web3/Solidity.

Work Experience

Senior Software Engineering Manager | SymphonyAI
Paris, France • September 2018 — Present

Led a data-driven initiative to optimize Retrieval-Augmented Generation (RAG) performance. Benchmarked multiple embedding models using Python (LangChain, Hugging Face) and Azure OpenAI against Oracle and FAISS storage platforms. Engineered a test suite of 14k+ chunks (technical docs and wiki data) to evaluate retrieval accuracy, resulting in the selection of the gte-large model for production.

Spearheaded the development of a new SymphonyAI SDK as an external product offering. Architected an ecosystem for partners including interactive tutorials, automated documentation, and a Custom AI Agent with specialized "AI Skills" designed to automate coding according to internal conventions and toolsets.

Architected and implemented the GOLD Application Observability Platform (GALOP) using OpenSearch, OpenTelemetry Collector, Data Prepper, and Nginx. The platform captures 1M spans and 8M metrics daily displayed across 10+ dashboards with corporate OIDC authentication. Automated the entire deployment via Docker Compose and Kubernetes with cross-platform compatibility for Splunk and Dynatrace.

Manage a cross-functional R&D team of 10 software engineers. Transformed the development culture by driving a wholesale migration to Azure DevOps, implementing rigorous CI/CD pipelines, and instituting automated unit testing (Karma, Jasmine, TestNG) and driving adoption of Katalon-based functional testing across the entire R&D that significantly reduced regressions and increased release velocity.

Software Development Lead | trueEx
New York City, NY • July 2014 — April 2018
Led a 5-person team building a real-time Interest Rate Swaps (**IRS**) trade processing system for our own CFTC-regulated exchange. Implemented **FRA**, Swaptions, and MAC financial products as well as architected APIs for trade ingestion via **FIX** connections and trading via **RFQ** across London and NYC trading sessions. Navigated high-pressure regulatory shifts by implementing ANNA ISIN integration for MiFID II compliance working closely with outside regulators.

Vice President | Citi
New York City, NY • May 2004 — July 2014
Developed proprietary FX trading algorithms based on close interaction with the trading desk. Performed extensive trading analysis in kdb+ using **q and MATLAB**. This was a result of first leading a team building out and managing high performance **kdb+ based infrastructure (dataseries and listeners) for trade capture and analysis.**

Designed a global data cache fabric using **GemFire** (100+ caches, 4MM+ records) for trading reference data (calendars, currencies, product symbols) with <1s global propagation across 6 data centers in North America, Europe and Asia.

Senior Software Engineer | Instinet
New York City, NY • September 1999 — May 2004
Designed **BookMark** (USPTO Serial # 76072768), a real-time web-based stock viewer using a then-novel "never-ending response" HTTP streaming concept. Developed a centralized Server Management console for monitoring application health and a Java Webstart-based employee directory.

Selected Projects & Publications
SKALA3MA (Gym & Competition Management): Developed a web app used by FSGT for 5 consecutive seasons to organize climbing competitions. It also ncludes gym route management, activity recording, and automated generation of wall-mounted route signage.

Line Text Scanner (Mobile ML): Google Play Android application utilizing Google MLKit. Engineered a custom "line priority" logic and skew-compensation algorithm to accurately extract data from tables.

Technical Writing: Published articles on **Hackernoon** regarding data visualization and API integration, specifically focusing on visualizing favorite artists over time using Python and Plotly.

Technical Profile
- **Systems & Data:** OpenSearch, Open Telemetry, Docker, Azure DevOps, KDB+, Kafka, PostgreSQL, Oracle
- **Languages & AI:** Java (Spring), Python (Pandas, Flask, Streamlit), q (Kx Systems), JavaScript (React/Vue), RAG, LangChain, Solidity.
- **Mobile & Security:** MS Intune/MDM, AppConfig, Android/iOS, OAuth2, JWT, Cordova, Capacitor.

Education & Coaching
- B.S. in Computer Science, Pace University, New York City.
- Mountain & Climbing Coach, FSGT Mountain and Climbing section.
- Interests: Rock climbing, mountain biking, snowboarding, windsurfing, diving, reading.